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ORIGINAL ARTICLE

A modified petal and stamen dimorphism interact to enhance pollen placement by a buzz-pollinated flower

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• **Background** Floral adaptations supposedly help pollen grains to cross the numerous barriers faced during their journey to stigmas. Stamen dimorphism and specialized petals, like the cucculus in the Cassieae tribe (Fabaceae), are commonly observed in flowers that offer only pollen as a resource for bee pollinators. Here, we experimentally investigated whether stamen dimorphism and the cucculus enhance pollen placement on the bee's body.
• **Methods** We used 3-D-printed bee models to apply artificial vibrations to the flowers of *Chamaechrista latistipula* with their cucculus deflected or maintained in its original position and their anther pores manipulated. After each simulated flower visit, we captured photographs of the artificial bee from four distinct angles. Employing digital imaging techniques, we documented the presence and location of pollen and stigma on the bee's body.
• **Key Results** Our findings reveal that the cucculus redistributes pollen grains on the bee's body. There is a remarkable increase in pollen density (~10-fold) on the lateral side adjacent to the cucculus, precisely where the stigma contacts the bee when the cucculus is unmanipulated. Furthermore, the cucculus also enhances pollen placement on the ventral region of the bee, indicating its additional function. The cucculus also increases the accuracy of pollen grains on the adjacent lateral region of the bee's body, irrespective of the pollen grains released by small or large anthers.
• **Conclusions** Floral specialized traits, such as modified petals and stamen dimorphism, can modify the fate of pollen grains and ultimately contribute to male reproductive performance in pollen flowers with poricidal anthers. The cucculus exhibits a dual role by promoting pollen placement in optimal regions for pollination and probably supporting pollen grains for bee feeding. These findings provide valuable insights into the adaptive significance of floral traits and their impact on the reproductive success of pollen flowers.

Key words: Buzz pollination, floral morphology, heteranthery, inaccuracy, male performance, poricidal anthers.

INTRODUCTION

One of the most notable examples of flower specialization is buzz-pollinated flowers that conceal their pollen grains within poricidal anthers (Vallejo-Marín *et al.*, 2010; Vallejo-Marín, 2019). These flowers are pollinated by female bees able to produce vibrations with their thoracic muscles, which in turn are mechanically transmitted to anthers, resulting in pollen release and a buzzing sound as a byproduct (Vogel, 1978; Buchmann, 1983). Buzz pollination independently evolved multiple times during angiosperm diversification, occurring in more than 22 000 extant plant species (Lunau, 2004; Vallejo-Marín and Russell, 2024; Russell *et al.*, 2024). As most buzz-pollinated flowers offer only pollen as a resource for buzzing bees, this system is thought to be under an intense 'pollen dilemma' – an evolutionary challenge of balancing pollen allocation between pollinator reward and plant fertilization (Westerkamp, 1996; Westerkamp, 1997).

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In addition to poricidal anthers, stamen dimorphism (i.e. more than one type of stamen within the same flower) is another common trait that emerged in buzz-pollinated flowers, potentially minimizing the pollen dilemma and increasing the accuracy of pollen transfer to the stigma (Renner, 1990; Luo *et al.*, 2009; Vallejo-Marín *et al.*, 2010; Cardoso *et al.*, 2018; Konzmann *et al.*, 2020). The functional role of stamen dimorphism (or heteranthery) was first hypothesized over a century ago (Müller, 1881, 1882, 1883), although experimental investigations into it were made only very recently (Luo *et al.*, 2009; Vallejo-Marín *et al.*, 2009, 2010; Paulino *et al.*, 2013; Paulino *et al.*, 2016; Li *et al.*, 2015; Mesquita-Neto *et al.*, 2017; Saab *et al.*, 2021). In buzz-pollinated flowers with stamen dimorphism, bees preferentially forage on the central conspicuous feeding anthers, while pollinating anthers are more external and less attractive to the bees, resulting in their pollen grains being more likely to be deposited on the stigma (Luo *et al.*, 2008; Vallejo-Marín *et al.*, 2009; Velloso *et al.*, 2018; Saab *et al.*, 2021). Both poricidal anthers and stamen dimorphism are understood to be a specialization for buzz pollination, which would increase the accuracy of pollen transfer and deposition, favouring the sexual reproduction of these plants.

Some groups of plants even have additional floral traits associated with the buzz-pollination process. A remarkable and recurrent feature of buzz-pollinated plants belonging to the tribe Cassieae (Fabaceae) (Rando *et al.*, 2024) is the presence of modified petals associated with the poricidal anthers (Marazzi and Endress, 2008; Almeida *et al.*, 2015*a*, *b*). In these flowers, pollen grains ejected from the poricidal anthers may have their trajectory modified by these petals during buzz pollination (Delgado-Salinas and Sousa-Sánchez, 1977; Westerkamp, 2004; Amorim *et al.*, 2017; Nogueira *et al.*, 2018), distributing the pollen grains on different regions of the bee's body relative to where the poricidal anthers would directly place them. This mechanism possibly increases pollen placement accuracy, which is fundamental to male fitness during buzz pollination. Despite the widespread occurrence of these modified petals and the dimorphic poricidal anthered stamens in flowers of the Cassieae tribe, the extent of their impact on pollen placement success has never been assessed.

In this study, we experimentally investigate the role of modified petals and stamen dimorphism in pollen placement in buzz-pollinated flowers with poricidal anthers. We conducted a greenhouse experiment using the legume plant species *Chamaecrista latistipula* with poricidal flowers bearing different types of stamens and a curved and rigid modified petal known as a 'cucculus' (Nogueira *et al.*, 2018; Rando *et al.*, 2021). This species belongs to the Cassieae tribe, one of the largest buzz-pollinated groups, in which stamen dimorphism and cucculus are prevalent. To explore the functionality of the distinct stamens and cucculus, we manipulated the flower structure, partially or totally obstructing the anther pores of the different stamens sets and bending or not bending the cucculus. Next, we applied artificial mechanical vibrations with a bee pollinator mimic on the flowers under different conditions. Then, we quantified pollen placement and the stigma position on the bee's body using digital imaging techniques. Within the scope of this experiment, we aim to address three key questions: (1) How does the cucculus impact the pollen deposition density in different regions of the bee's body? (2) How do different stamens influence the pollen density on the lateral regions of the bee's body adjacent to the cucculus? (3) To what extent do the cucculus and stamen dimorphism reduce inaccurate pollen placement on the bee's body, and how do they contribute to male reproductive performance in these flowers? We hypothesize that the cucculus and large stamens increase pollen density and placement accuracy on the bees' bodies, potentially increasing male reproductive performance in buzz-pollinated flowers.

MATERIALS AND METHODS

Flower species and study site

Chamaecrista latistipula (Fig. 1) is a shrubby species that exhibits wide distribution in the Brazilian Cerrado ecosystem (Nogueira *et al.*, 2018). During its flowering season between October and March (Fidalgo *et al.*, 2018), this plant species produces vibrant yellow flowers that attract hymenopteran and coleopteran visitors (Ruhren, 2003; Camargo and Miotto, 2004). The flowers of *C. latistipula* last one light day and are characterized by having five petals, with one of them forming a distinctive and specialized structure known as the cucculus. The cucculus is the innermost lateral petal, concave, slightly thicker, and more robust than other petals in *C. latistipula* flowers (Nogueira *et al.*, 2018). Additionally, these flowers display stamen dimorphism, with three large stamens and seven smaller ones that are centrally positioned. Unlike other species within Cassiinae, where larger and smaller stamens occupy different positions within the flower and have different colours for bee pollinators (e.g. Saab *et al.*, 2021), the stamens of *C. latistipula* are similar in colour and positioned centrally in the flower, differing only in size. Moreover, the seven smaller stamens contribute two-thirds of the pollen grains in the flower, while the remaining third is distributed among the three larger stamens (A. Nogueira, unpubl. data). Therefore, the division of labour in the *C. latistipula* flower is likely much less pronounced, and it becomes challenging to assess beforehand the significance of stamen size dimorphism in directing pollen to the bee's body, given that their arrangement can be influenced by the modified petal, the cucculus (Nogueira *et al.*, 2018).

Chamaecrista latistipula plants also exhibit non-reciprocal monomorphic enantiostyly, producing mirror-image flowers within the same plant. Half of the flowers produced by an individual *C. latistipula* plant have styles deflected to the right, while the other half have styles deflected to the left. Although the positioning of anthers relative to the styles is not reciprocal, the cucculus always shows deflection in the opposite direction compared with the style within the same flower.

The experiment took place in October 2021 at the Federal University of ABC in São Bernardo do Campo, São Paulo, Brazil, coinciding with the peak of the flowering phase of *C. latistipula*. There, an experimental population of *C. latistipula*, initially transplanted from a natural population in Mairiporã, São Paulo (latitude −23.320177, longitude −46.584018), was used as a seed source. Seeds obtained from these specimens were sown in 11-L pots, containing a layer of expanded clay at the bottom and filled with a homogenized substrate comprising

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FIG. 1. (A) *Chamaecrista latistipula* flower being visited by a buzzing bee, *Bombus morio*. (B) Front view of *C. latistipula* flower with unmanipulated cucculus. (C) Front view of *C. latistipula* with deflected cucculus. (D) 3-D artificial bee model attached to a wooden rod connected to the vibration transductor speaker featuring a vibrating metal plate. *Chamaecrista latistipula* flowers are characterized by five bright yellow petals, one of them forming a distinctive curved structure known as the cucculus. Also, the flower displays stamen dimorphism, with three large stamens and seven smaller ones that are centrally positioned and bear poricidal anthers.

sand and vegetal soil. Subsequently, the seeds were planted superficially at 0.5–1 cm below the soil surface. The plants were cultivated in a greenhouse at the university and irrigated using automatic irrigators positioned near the pots for 2 years before the experiments.

Artificial vibrations

We constructed a 3-D artificial bee model to examine the impact of the cucculus and stamen dimorphism on pollen density and inaccuracy when placed on a bee's body. Artificial bees were used to keep the bee size and vibrations standardized and to avoid killing bees. The model was created using a 3-D printer (Anycubic Photon Mono X) and black resin material (Anycubic Black, viscosity at 25 °C of 552 mPa s). Although it was impossible to print bee hairs, the pollen grains visually attached well to the resin material, making pollen placement comparable among treatments. The dimensions of the artificial bee's body were determined based on the length and intertegular distance of *Bombus morio*, a key pollinator of *C. latistipula* flowers (Fidalgo *et al.*, 2018). This artificial bee model was affixed to a pin, which was attached to a wooden rod measuring 6 cm in height and 4 mm in diameter. The rod and the longitudinal axis of the bee's body formed a 45° angle. Subsequently, the wooden rod was connected to a vibration transducer speaker featuring a vibrating metal plate (Youts Globe Super Speaker).

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This speaker transduced signals designed to simulate the vibrations produced by bees during their flower visitation.

The artificial vibrational signals were generated on a computer using Audacity software (www.audacityteam.org). We synthesized sine waves with a frequency of 300 Hz, a relative amplitude of 1, a sampling rate of 44.1 kHz and a phase $\varphi = 0°$. These signals were designed to mimic the vibrations produced by buzzing bees during their visitation to *C. latistipula* flowers, drawing upon different systems described in previous studies (de Luca *et al.*, 2013; Arroyo-Correa *et al.*, 2019; Brito *et al.*, 2020; Pritchard and Vallejo-Marín, 2020). The artificial vibration consisted of eleven 1-s pulses separated by 200 ms of silence. The transducer speaker's absolute amplitude velocity was directly adjusted to achieve 100 mm s^{-1}.

We used a laser Doppler vibrometer (PDV-100, Polytec) for system calibration, ensuring the signals reproduced as vibration in the artificial bees remained consistent during the experiment. The laser beam was directed parallel to the main displacement axis of the transducer, continuously targeting the artificial bee's head. The vibrometer settings included a low-pass filter of 5 Hz, no high-pass filter, and a laser speed of 500 mm s^{-1}. The calibration vibrometer was connected to a secondary computer, enabling real-time monitoring of the artificial bee's vibration using the VibSoft-20 software in the oscilloscope function. Supplementary Data Fig. S1 shows the details of the experimental setup.

Experiment and image treatments

We conducted a 2×3 factorial experiment manipulating the cucculus (two levels) and anther pores (three levels) to investigate the impact of the cucculus and stamen dimorphism on pollen placement. Six different conditions were created manipulating both structures as follows: (1) flowers with unmanipulated cucculus and both types of anthers unobstructed (complete control); (2) flowers with unmanipulated cucculus and unobstructed small anthers; (3) flowers with unmanipulated cucculus and unobstructed large anthers; (4) flowers with deflected cucculus and both types of anthers unobstructed; (5) flowers with deflected cucculus and unobstructed small anthers; and (6) flowers with deflected cucculus and unobstructed large anthers. We performed 36 replicates on each treatment (see Supplementary Data Table S1 for replication statement), using a new fresh flower for each replicate. The number of replicates analysed may differ from this original number because some data were impossible to recover due to a lack of contrast between the pollen grains and bee resin, spoiling a proper binarization during image processing (see below, Supplementary Data Fig. S3).

To prevent any alteration of floral biomechanical properties resulting from mass loss or lack of contact with the artificial bee, we manipulated the cucculus by deflecting it instead of removing it from the flowers. The obstruction of large or small anthers was achieved using methacrylate superglue (Locite Super Bonder). We also placed an equivalent drop of superglue on the unobstructed anthers to control for any potential effect of this manipulation. Subsequently, the treated flowers were carefully detached from the plants and manually secured to the artificial bee coupled to the vibration transducer. Following the vibration, we marked the position of the stigmas on the side of the bee's body opposite the cucculus using fluorescent powder (Pigmento Fluorescente em Pó Rosa, Hazzin). The artificial bee was then decoupled from the vibration transducer and photographed (using either a Canon EOS Rebel SL3 camera with a 100-mm macro lens or a Samsung NX300 camera) from four different angles, emphasizing its lateral region adjacent to the cucculus, dorsal region, lateral region opposite to the cucculus, and ventral region, while placed on a black foam stage (Supplementary Data Fig. S2). Due to the variation in sensor characteristics between the two cameras, images captured with the Samsung NX300 were resampled to match the resolution of those taken with the Canon EOS Rebel SL3. We used the nearest neighbour interpolation technique, which ensures uniform spatial characteristics across all datasets, preserving the original pixel values.

We utilized a series of image-processing steps to analyse the distribution of pollen grains in different regions of the bee's body. Initially, the original images were subjected to binarization (Supplementary Data Fig. S2) using GIMP (GNU Image Manipulation Program). In this process, pixels with one value represented pollen-occupied areas, while pixels with a zero value represented non-pollen-occupied areas. Following binarization, each image was manually aligned on Cartesian axes to accurately represent the bee's size. Alignment was achieved using two tying points, i.e. reference location, at each lateral extremity of the bee's body. This procedure allowed us to overlap all images in a 2-D space. To ensure accurate overlapping, we randomly selected one image per treatment as a model for cropping the other images. This step was accomplished using the image transformation tool for polygons, whereby the bee body was delimited in each treatment. The smoothed polygon was obtained through edge smoothing utilizing the polynomial approximation algorithm with an exponential kernel and a smoothing tolerance of 1 mm. The remaining images were subsequently cropped based on the smoothed polygon.

Next, we transformed each pixel occupied by pollen into a point and extracted the corresponding spatial x and y coordinates of each point. We used this information to produce heat maps of pollen placement for each treatment (Supplementary Data Fig. S3). We also estimated pollen density on the bee's body by dividing the number of occupied pixels in each image by the total number of pixels available (i.e. the number of occupied pixels plus the number of unoccupied pixels). The image processing tasks for resampling the images, extracting pollen coordinates and estimating pollen density were performed using the Resample, Geoprocessing, Raster to Polygon, Raster to Point, Smooth Polygon, and Add XY Coordinates tools within ArcGIS Pro version 3.0.1.

Estimation of the inaccuracy components in the bee body

In our study, we estimated two components of the realized inaccuracy to assess the impact of the cucculus and stamen dimorphism on the position of pollen grains on the bee body relative to the position where the stigma touches. Inaccuracy is defined as the population deviation from a given optimum, being described by (1) the maladaptive bias, i.e. the mean departure from the optimum, and (2) the imprecision, i.e. the

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intrapopulation variation (*sensu*). To estimate the mean departure from the optimum, we utilized the coordinates of each pollen-occupied pixel in the digitized images of the bee body. Subsequently, we determined the distance between the average position of the pollen and the stigma optimum (i.e. the stigmas' average coordinate where they touch the bee body). It is important to note that, as *C. latistipula* has enantiostyly with two floral morphs, the population has two stigma optima on the bee's body. Therefore, the departure from the optimum was calculated considering the corresponding bee lateral region in each case. The distance measurement represents the first component of the inaccuracy, i.e. the maladaptive bias of pollen following each bee visit. In addition, we computed the sum of the standard deviations of the x and y coordinates of the pollen positions to estimate the second component of the inaccuracy, the realized intrapopulation variation in pollen placement. We chose to use the standard deviation rather than the variance proposed by due to our dataset's absence of negative values and to achieve better model fit.

Statistical analysis

The data analysis was structured following the three key questions stated in the last paragraph of the Introduction.

Firstly, to investigate how the cucculus impacts the pollen deposition density in different regions of the bee's body, we built a binomial generalized linear mixed model in which the response variable was the total number of pixels occupied by pollen on the bee's body combined with the number of unoccupied pixels. The explanatory variables were categorical and included the condition of the cucculus (deflected or unmanipulated), the region of the bee's body (lateral adjacent to the cucculus, dorsal, lateral opposite the cucculus, or ventral), as well as their interaction. For this analysis, we used data from treatments 1 and 4 (i.e. control flowers and flowers with deflected cucculus and both types of anthers unobstructed, respectively), totalling 252 analysed images. Plant identity was considered a random factor to control for morphological variation due to genetic basis, and a binomial error distribution with the logit link function was employed. After fitting the model, a type III ANOVA was conducted to determine the significance of each explanatory variable once we expected an interaction effect between the explanatory variables. Additionally, we compared the marginal means across different levels of the explanatory variable.

Secondly, to investigate the influence of different stamens on the pollen density placed on the lateral region of the bee's body adjacent to the cucculus, we employed a binomial generalized linear mixed model, in which the response variable was also the total number of pixels occupied by pollen on the bee's body, combined with the number of unoccupied pixels. The explanatory variable considered in this analysis was categorical in three levels: both types of anthers unobstructed (control), small or large anthers, both unobstructed. Here, we used data from treatments 1, 2 and 3 (i.e. all treatments involving unmanipulated cucculus, varying only the anthers condition), totalling 154 analysed images. Plant identity was also included as a random factor, and a binomial error distribution with the logit link function was applied. Following model adjustment, we conducted a type II variance analysis to assess each explanatory variable's significance. Furthermore, we compared the marginal means across different levels of the explanatory variable. In this second analysis, we did not include data on bee vibrations in flowers with deflected cucculus. We conducted these two analyses separately rather than running a full model accounting for all six treatments because we previously knew that the other treatments produced many low or even zero pollen densities (Supplementary Data Fig. S3).

Thirdly, to investigate the roles of the cucculus and stamen dimorphism in the realized inaccuracy of pollen placement on the lateral region of the bee's body, we estimated the two components of inaccuracy and used them as response variables in two different generalized mixed models. The explanatory variables used in the models were the condition of the cucculus (deflected or unmanipulated), the condition of the anthers (both types of anthers unobstructed, small anthers unobstructed, or large anthers unobstructed), and their interaction. Here, we considered all treatments but only the lateral side adjacent to the *cucullus* because this is the region where the stigma touches the bee's body and showed the highest density of pollen grains. We considered a total of 182 images for each analysis. Plant identity was included as a random factor, and a Gaussian error distribution was assumed. We conducted a type III variance analysis to evaluate each explanatory variable's significance. Additionally, we estimated the marginal means for posterior comparisons.

The data analyses were conducted using various packages in the R version 4.1.2 environment (http://www.r-project.org/). The specific packages used include ggplot2 (Wickham, 2016) for data visualization, glmmTMB (Brooks *et al.*, 2017) for fitting generalized linear mixed models, emmeans (Russell *et al.*, 2023) for estimating marginal means, car (Fox and Weisberg, 2019) for additional analyses and the DHARMa package (Hartig, 2024) for checking residual dispersion around the fitted models. We responsibly used ChatGPT to enhance the manuscript's clarity, coherence and overall readability.

RESULTS

Effect of the cucculus on pollen deposition density on the bee's body

The cucculus affects the placement of pollen grains on the body of artificial bees (Supplementary Data Fig. S3; Supplementary Data Table S2; Supplementary Data Video S1). Bees vibrated on flowers with an unmanipulated cucculus showed a higher density of pollen grains placed on their bodies than bees vibrated in flowers with a deflected cucculus (Fig. 2; Supplementary Data Table S2). However, this relationship depends on which region of the bee's body is considered (Fig. 3; Supplementary Data Table S2; $\chi^2 = 13021.76$; d.f. = 3; $P < 0.01$). On the side of the bee body adjacent to the cucculus, unmanipulated flowers placed almost 10 times more pollen grains than flowers with deflected *cucculus* (Fig. 3; Supplementary Data Table S3; ratio = 9.88; $z = 254.88$; $P < 0.01$). In addition, flowers with an unmanipulated cucculus also increased by 2.3 times the density of pollen grains placed on the ventral region of the body of artificial bees (Fig. 3; Supplementary Data Table S3; ratio = 2.31; $z = 79.09$;

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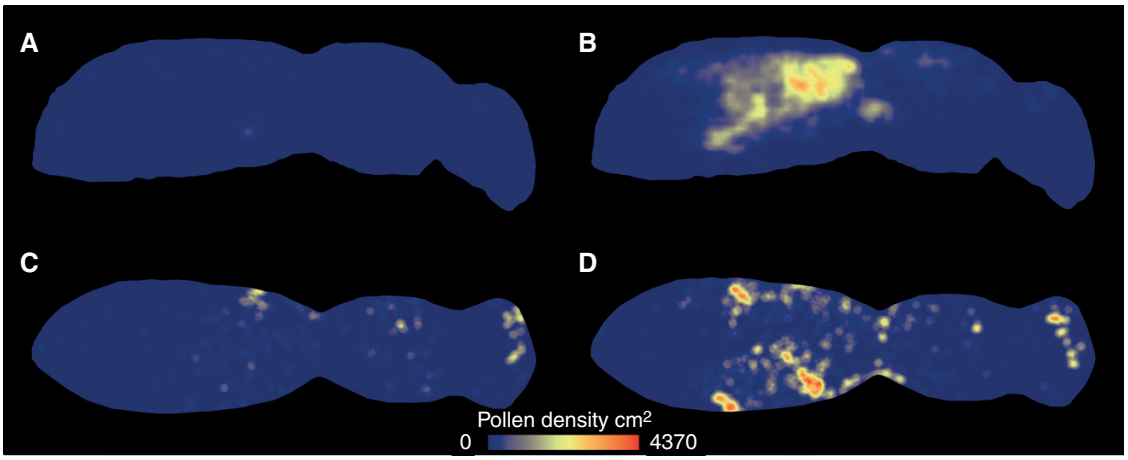


FIG. 2. Pollen deposition density on the bee's body. Increasingly warm colours highlight higher pollen densities until red. (A) and (B) are images of the lateral bee's side adjacent to the cucculus deflected and unmanipulated, respectively. (C) and (D) are images of the ventral region of the bee's body in flowers with cucculus deflected and unmanipulated, respectively.

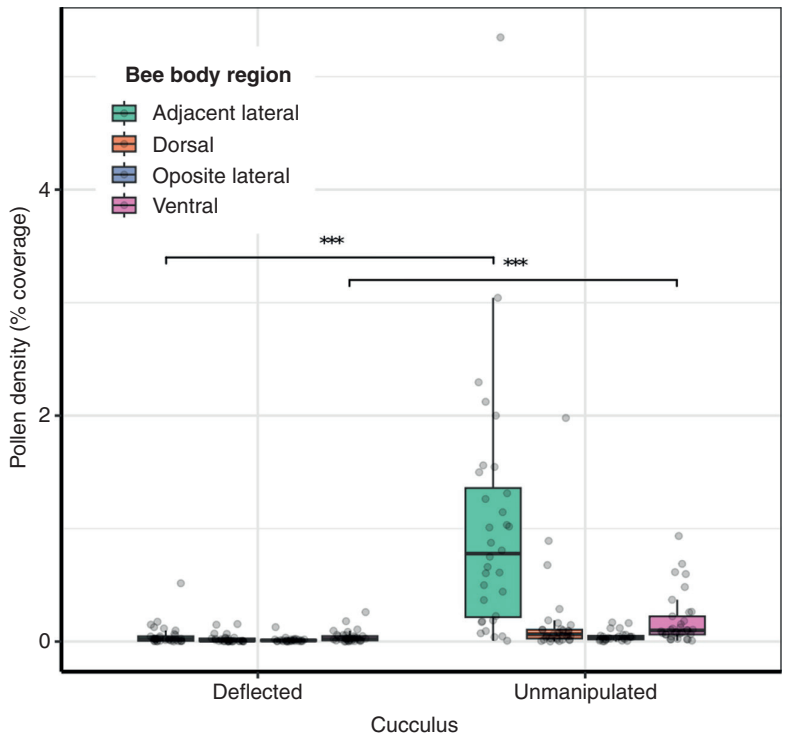


FIG. 3. Pollen deposition density on different regions of the artificial bee's body. Pollen density was higher when the cucculus was left unmanipulated in both adjacent lateral and ventral regions. The artificial bee was built in a 3-D printer and coupled to a vibration transducer to simulate bee visits. Pollen density estimation was done using digital imaging techniques. *** indicates a significant contrast between estimated marginal means at a 95 % confidence interval.

$P < 0.01$). Notably, little to no pollen reached the bee's dorsal and opposite lateral regions in our simulations, regardless of cucculus manipulation (Supplementary Data Fig. S3).

Effect of stamen dimorphism on pollen density on the bee lateral side adjacent to the cucculus

The pollen density on the bee side adjacent to the unmanipulated cucculus depended on stamen dimorphism (Fig. 4; Supplementary Data Table S4; $\chi^2 = 50055.00$; d.f. = 2; $P < 0.01$). Flowers with all unobstructed anthers placed up to almost 3 times more pollen grains than other anther treatments (Fig. 4; Supplementary Data Table S5; $ratio_{both\text{-}only\ large} = 4.55$; $z = 194.18$; $P < 0.01$; $ratio_{both\text{-}only\ small} = 2.85$; $z = 155.70$; $P < 0.01$). Moreover, the pollen density deposited by small anthers was almost twice that of large anthers (Fig. 4; Supplementary Data Table S6; $ratio_{only\ small—only\ large} = 1.59$; $z = 52.09$; $P < 0.01$).

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Effect of the cucculus and stamen dimorphism on the inaccuracy components of pollen grains

The two inaccuracy components were affected differently by the cucculus and stamen dimorphism. Firstly, the mean departure from the optimum was not modified by either the cucculus manipulation or the obstruction of anther types (Fig. 5A; Supplementary Data Table S6; $\chi^2 = 1.29$; d.f. = 2; $P > 0.05$). Therefore, the pollen positioning relative to the stigma does not depend on the cucculus or specifically on any anther type, although both strongly influence the pollen grain density.

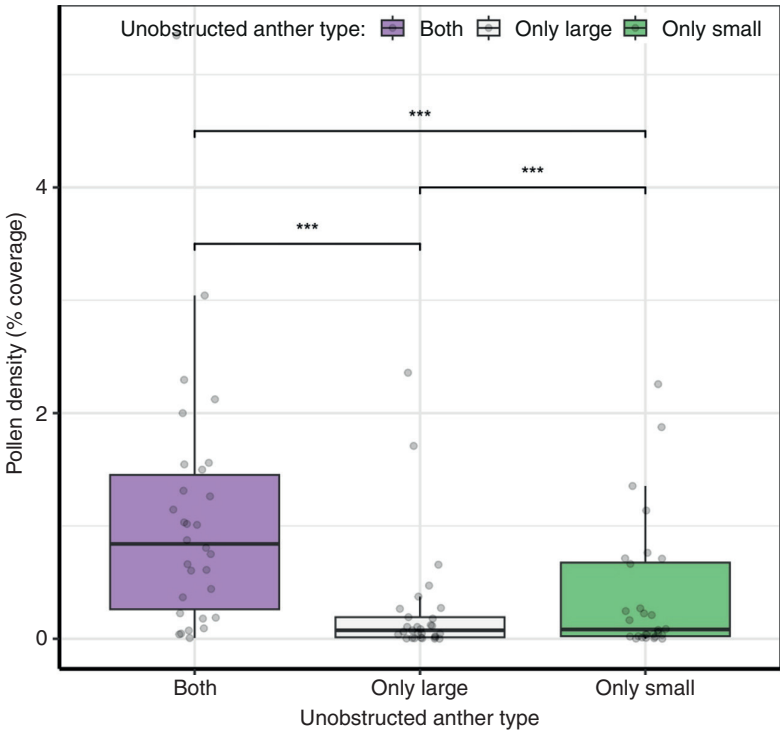


FIG. 4. Pollen density deposited on the bee body side adjacent to the cucculus in flowers with unmanipulated cucculus under three conditions of anther pore manipulation. Both anthers were left unobstructed or the anthers of either large or small stamens were obstructed using superglue. Pollen density was higher when both anthers were unobstructed. The artificial bee used to vibrate the flowers was built in a 3-D printer and coupled to a vibration transducer to simulate bee visits. *** indicates a significant contrast between estimated marginal means at a 95 % confidence interval.

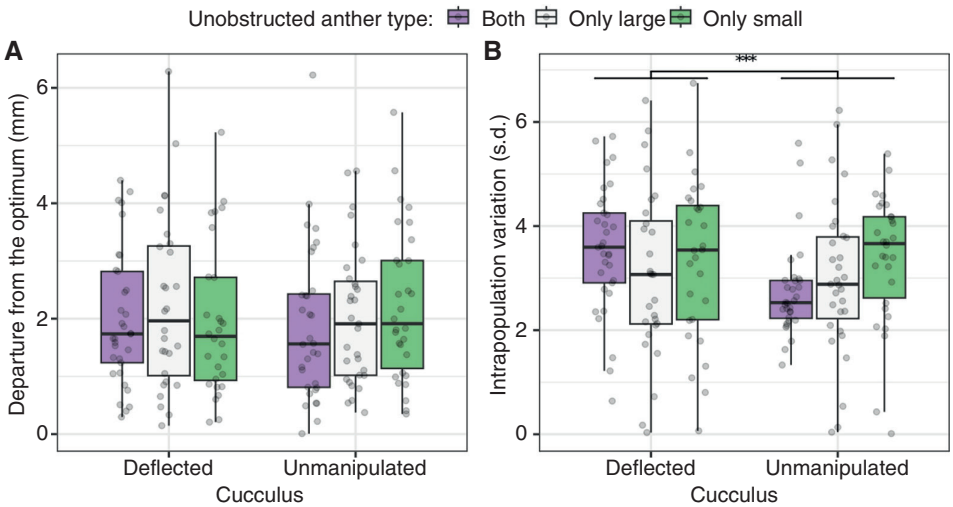


FIG. 5. Components of the realized inaccuracy of pollen grains on the lateral region of the bee body adjacent to the cucculus. (A) Departure from the optimum (stigma position) across the cucculus and anther obstruction conditions. (B) Intrapopulation variation (standard deviation) across the cucculus and anther obstruction conditions. Flowers had the cucculus unmanipulated or manipulated and the two types of anthers unobstructed, only the small anthers obstructed or only the large anthers obstructed. Both departure from the optimum and intrapopulation variance were estimated from the coordinates of each occupied pixel in the digitized images. *** indicates a significant contrast between estimated marginal means at a 95 % confidence interval.

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On the other hand, the intrapopulation variation (standard deviation) was impacted by manipulating the cucculus but not by anther obstruction. In this case, unmanipulated flowers placed more clumped pollen loads than flowers with deflected cucculus (Fig. 5B; Supplementary Data Table S7; $\chi^2 = 239.61$; d.f. = 1; $P < 0.05$), i.e. the deflection of the cucculus increased pollen placement imprecision by ~10 % on the bee's body.

DISCUSSION

In this study we tracked the position of pollen grains on an artificial bee body, which allowed us to investigate the role of stamen dimorphism and the specialized petal called the cucculus of *C. latistipula* flowers in pollen placement, an essential step in the pollen's journey towards the stigmas. We hypothesized that the cucculus and large stamens increase pollen density and placement accuracy on the bees' bodies. In fact, our findings revealed that the cucculus enhances pollen deposition on the bee body region adjacent to it and on the bee's venter. However, contrary to our expectations, small stamens contribute more pollen grains than large ones, always mediated by the cucculus, indicating a deviation from the typical function of stamen dimorphism in flowers with poricidal anthers. The contribution of smaller stamens to the pollination process explains the similar pollen viability across stamen types (Nogueira *et al.*, 2018), in contrast to what has been previously reported in other studies (e.g. Saab *et al.*, 2021). Additionally, the cucculus also influences pollen placement accuracy, reducing pollen placement imprecision on the bee's body and guiding the pollen towards more favourable regions for reaching the stigmas of *C. latistipula*. Although 'ricochet pollination' has been documented in other flowers with a cucculus (Westerkamp, 2004; Amorim *et al.*, 2017), this study provides the first quantitative estimation of its impact on pollen placement by both stamen types, including the smaller ones. This process is facilitated by modified petals, which, in turn, enhance the flower's male function.

During buzz pollination in pollen flowers, both pollinator and plant can derive benefits based on the specific regions of the bee's body where pollen is placed. Generally, pollen grains placed in easily accessible body regions can be collected through grooming behaviour, ensuring resources for the bee's offspring (Thorp, 1979, 2000). On the other hand, pollen grains deposited on safe sites, which are located outside of grooming regions and in closer proximity to where the stigmas make contact, have a higher likelihood of effectively accomplishing pollination and subsequent fertilization of the ovules (Buchmann, 1983; Thorp, 2000; Koch *et al.*, 2017; Tong and Huang, 2017). In such buzz-pollinated flowers, the division of the pollen load into different portions is primarily facilitated by different stamens within the flower – the stamen dimorphism that evolved in numerous lineages of pollen flowers (Vallejo-Marín *et al.*, 2010; Melo *et al.*, 2021). However, our findings demonstrate that the unequal distribution of pollen grains in these safe and unsafe sites on the bee's body is primarily mediated by the modified petal cucculus in *C. latistipula*. By enhancing pollen deposition on the bee's lateral body region adjacent to the *cucullus*, this petal increases pollen density on a safe site – where bees have limited access, but where stigmas of enantiostylous flowers frequently make contact during visits (Koch *et al.*, 2017; Tong and Huang, 2017). Thus, by facilitating such a distribution of the pollen load, the cucculus primarily serves a male function during the pollination process. Besides its reproductive role, the cucculus also serves a feeding function, as evidenced by the higher density of pollen in the ventral region of the bee's body when the cucculus was unaltered. These results highlight that the division of labour achieved solely through stamen dimorphism in most pollen flowers can be reinforced by specialized petals.

The evolution of offering only pollen as a floral reward for pollinators has been correlated with the existence of poricidal anthers and stamen dimorphism throughout the evolutionary history of angiosperms (Vallejo-Marín *et al.*, 2010; Russell *et al.*, 2024). In most flowers exhibiting these traits, larger stamens typically carry higher viable pollen loads, and their pollen grains are expected to have a greater chance of reaching the stigmas during pollination (Maia *et al.*, 2019; Velloso *et al.*, 2018; Dellinger *et al.*, 2019; Brito *et al.*, 2021; Saab *et al.*, 2021; Trevisan *et al.*, 2023). However, recent studies have challenged this well-established function of different stamens (Kay *et al.*, 2020; Konzmann *et al.*, 2020; Telles *et al.*, 2020; Dellinger *et al.*, 2021; Trevisan *et al.*, 2023). For instance, in *Clarkia unguiculata* (Onagraceae), small stamens produce pollen that outperforms large stamens in both stigma and style penetration (Peach and Mazer, 2019). A similar pattern is observed in *Pterolepis glomerata* (Melastomataceae), in which small stamens produce more viable pollen and release more pollen grains during a single bee visit than large stamens (Telles *et al.*, 2020). Interestingly, in the case of *C. latistipula*, the role of the cucculus contradicts our initial expectations, as the small stamens placed more pollen grains than the large stamens on the lateral region of the bee's body. Thus, specialized petals in buzz-pollinated flowers can redirect pollen grains produced by the smaller stamens, which would primarily serve as a food source for bees, towards pollination. In fact, the anthers of the seven small stamens in this species seem to produce more pollen grains, ~64 % of the total pollen load per flower (A. Nogueira, unpubl. data), suggesting that the difference observed in pollen deposition may be just proportional to the total pollen produced by each stamen type. The proportional contribution of stamen types for pollination reinforces that the functional role of stamen types may be labile or even absent in some flowers with stamen dimorphism (Konzmann *et al.*, 2020).

Pollen flowers, including *C. latistipula*, often exhibit enantiostyly, a floral system where the pistil is deflected to one side of the floral symmetry axis (Fidalgo *et al.*, 2018). This deflection is commonly accompanied by the reciprocal positioning of stamens, corresponding to the stigma position of other flowers in the population, reducing the likelihood of self-pollination and promoting cross-pollination (Webb and Lloyd, 1986; Braga *et al.*, 2022). Like heterostylous systems, enantiostyly can be considered a form of reciprocal herkogamy (Cardoso *et al.*, 2018). In previous studies of heterostylous flowers, measures of inaccuracy have been used to assess the reciprocity between stamens and stigmas (Matias *et al.*, 2020; Raupp *et al.*, 2020; Trevizan *et al.*, 2021; Braga *et al.*, 2022). However, in our study we introduced a novel approach using both maladaptive bias and imprecision to estimate the impact of the cucculus and stamen dimorphism on the position of pollen grains relative to the stigmas on the bee's body. Our findings

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demonstrate that the cucculus reduces the realized inaccuracy of pollen grains by decreasing their imprecision around the optimal position for stigmatic contact.

In those cases, specialized petals like the cucculus can enhance the deposition of pollen grains in optimal locations for effective pollination in enantiostylous flowers. However, it is important to interpret the effect of cucculus manipulation on pollen placement imprecision with caution, as the observed impact was relatively low. Additionally, it should be noted that in natural settings pollen grains deposited on the bodies of real pollinators, especially bees, may form 3-D layers that vary in their exposure to stigmatic contact (Price and Waser, 1982; Lertzman and Gass, 1983; Marcelo *et al.*, 2022; Moir and Anderson, 2023). Since our study relied on 2-D photographs, our analysis did not consider the third dimension of pollen placement. The cucculus may decrease the realized inaccuracy of pollen grains by layering them on the bodies of actual bees. However, this hypothesis remains untested and would require further investigation.

CONCLUSIONS

Our study on *C. latistipula* revealed that the cucculus causes pollen grains to ricochet and be placed on the side adjacent to it and in the ventral region of the bee's body. For the first time, we quantitatively estimated the redistribution and increased placement of pollen grains on the bee's body, resulting from the presence of the cucculus. Furthermore, our findings indicate that pollen grains from both anther types, although especially from small anthers, can be directed towards plant reproductive purposes through their interaction with these specialized petals. By placing pollen grains on safe sites of voracious pollen collectors, commonly modified petals, such as the ones of the Cassieae tribe, plays a significant role in increasing the efficiency of pollen placement and, ultimately, pollen transfer. Our results highlighted the functional mechanism by which the modified petals enhance pollen placement on safe sites for plant sexual reproduction. In these cases, such specialized petals may have evolved mainly by increasing flowers' male reproductive performance.

SUPPLEMENTARY DATA

Supplementary data are available at *Annals of Botany* online and consist of the following. Table S1: ANOVA type III table to compare the effect of *cocculus* manipulation on pollen placement by *C. latistipula* in different bee body regions. Table S2: estimated marginal means and contrast between treatments (manipulated or unmanipulated cucculus) based on the adjusted model created to compare the effect of cucculus manipulation on pollen placement by *C. latistipula* in different bee body regions. Table S3: estimated marginal means and contrasts between different bee body regions (lateral adjacent to the cucculus, dorsal, lateral opposite the cucculus, or ventral) based on the adjusted model created to compare the effect of cucculus manipulation on pollen placement by *C. latistipula* in different bee body regions. Table S4: ANOVA type II table to assess the effect of stamen dimorphism on pollen placement by *C. latistipula* flowers on the bee lateral side adjacent to the cucculus. Table S5: estimated marginal means and contrasts between different anther treatments (both anthers unobstructed, only large anthers unobstructed, or only small anthers unobstructed) based on the adjusted model created to compare the effect of stamen dimorphism on pollen placement by *C. latistipula* in the lateral region of the bee body adjacent to the cucculus. Table S6: ANOVA type III table to compare the effect of cucculus manipulation and stamen dimorphism on the maladaptive bias of pollen placement by *C. latistipula* in the lateral region of the bee's body adjacent to the cucculus. Table S7: ANOVA type III table to compare the effect of cucculus manipulation and stamen dimorphism on the imprecision of pollen placement by *C. latistipula* in the lateral region of the bee's body adjacent to the cucculus. Figure S1: experimental setup. Figure S2: following the vibration of each flower, the artificial bee was decoupled from the vibration transducer and was photographed from four different angles, emphasizing its lateral region adjacent to the cucculus (A), dorsal region (C), lateral region opposite to the cucculus (E) and ventral region (G), while placed on a black foam stage. Figure S3: density of pollen placed on the four regions of the bee's body. Video S1: video showing the effect of the cucculus on pollen deposition density on the bee's body.

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DATA AVAILABILITY

All the data used in this research will be archived correctly in the Dryad repository (https://datadryad.org/stash) upon acceptance for publication.

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